Emergent Socio-Economic Dynamics in Autonomous Large Language Model Agents within Resource-Constrained Spatial Environments

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***Abstract*—This electronic document is a "live" template and already defines the components of your paper [title, text, heads, etc.] in its style sheet. **CRITICAL: Do Not Use Symbols, Special Characters, Footnotes, or Math in Paper Title or Abstract.* (*Abstract*)**

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I. Introduction

Creating computational environments that convincingly simulate human sociological and economic behaviors has been a persistent challenge in artificial intelligence and interactive system design. Historically, commercial life simulation environments, such as The Sims, have relied heavily on finite state machines, utility based artificial intelligence, and predefined behavior trees to govern Non-Player Characters [1]. While these deterministic architectures are highly optimized for executing routine tasks with minimal computational overhead, they are fundamentally constrained by rigid and pre-programmed rules. The NPCs in these systems lack episodic memory, cognitive reflection, and the capacity to generate unscripted social structures beyond the explicit parameters set by the developers [1].

Recent advancements in natural language processing have introduced a paradigm shift from solitary intelligence to collaborative and autonomous entities [2]. The integration of Large Language Models as the cognitive engines for software agents has birthed the concept of generative agents [2]. Unlike traditional NPCs, generative agents can perceive their environment, store observations in a vectorized memory stream, reflect on past experiences to form higher level insights, and autonomously plan future actions in a humanly manner. This cognitive architecture allows for dynamic adaptability and complex reasoning, establishing a new frontier for Multi Agent Systems [3].

However, scaling these LLM based cognitive architectures into real time multi agent spatial environments introduces severe technical bottlenecks. The primary challenges include high inference latency, context window saturation, and the computational cost of continuous memory retrieval [3]. When dozens of autonomous agents constantly query an LLM to determine their next micro action, the system experiences exponential performance degradation. Consequently, researchers must design highly optimized retrieval mechanisms and memory clustering algorithms to ensure that agents can react to their environment in real time without exceeding the computational limits of local hardware [4].

Beyond the technical constraints, there is a critical contextual gap in understanding the emergent behaviors of generative agents when subjected to resource scarcity and survival pressures [5]. Most existing frameworks evaluate agent coherence in benign and resource abundant scenarios, where agents merely simulate daily routines or engage in casual conversations. Foundational sociological and economic theories dictate that complex societal structures, such as bartering systems, trade alliances, and hierarchical governance, are born out of necessity when resources are limited [5]. Evaluating LLM agents solely in environments devoid of physical needs significantly restricts our understanding of their emergent evolutionary capabilities [6].

To bridge this gap, this paper proposes a novel computational architecture that embeds autonomous LLM agents within a highly constrained spatial sandbox. By injecting critical survival parameters specifically energy depletion, hunger, and inventory management this research forces agents into scenarios where self-preservation requires continuous social and economic interaction. The proposed simulation engine logically decouples the spatial physics loop from the heavy cognitive LLM processing, ensuring synchronous execution and stable tracking of agent states over extended periods without causing system timeouts [6].

The primary contribution of this study is threefold. First, we introduce a robust and low latency game loop engine capable of managing the spatial and temporal states of multiple LLM agents synchronously. Second, we propose an optimized memory retrieval architecture that reduces LLM inference latency during high frequency agent interactions. Third, we provide a quantitative sociological analysis of the unprogrammed behaviors of the agents, evaluating whether isolated entities will inherently develop collaborative economic systems or default to competitive monopolistic behaviors under extreme scarcity [7].

The remainder of this paper is structured as follows. Section II reviews related work in traditional game AI, generative agents, and simulated economics. Section III details the system architecture of the spatial engine and the cognitive memory stream. Section IV outlines the experimental methodology and the injected survival parameters. Section V presents the quantitative analysis of the emergent economic behaviors. Finally, Section VI concludes the paper and discusses avenues for future research [7].

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II. Related Work

For decades, the behavioral logic of Non-Player Characters in life simulation environments has been governed by deterministic architectures, primarily Finite State Machines and Behavior Trees [8]. These rule-based systems evaluate environmental conditions through predefined utility scoring. Mathematically, the utility U of executing an action a given the current state s is explicitly formulated as a weighted linear combination of satisfied parameters:

$$U(a|s) = \sum_{i=1}^{n} w_i \cdot f_i(s) \tag{1}$$

where w_i represents the predefined weight of a specific survival parameter (e.g., energy or hunger) and $f_i(s)$ denotes the hardcoded evaluation function for that condition [8]. The NPC strictly selects the action $\arg max_a \, U(a|s)$. While this approach provides a predictable and highly controllable framework that minimizes processing overhead, it suffers from severe cognitive limitations [9]. Traditional NPCs lack episodic memory and zero-shot reasoning capabilities, making them fundamentally incapable of generating unscripted emergent behaviors or adapting to unforeseen socioeconomic scenarios [10].

To overcome these deterministic limitations, the integration of Large Language Models has introduced cognitive architectures capable of natural language planning [11]. Generative agents utilize LLMs as their central cognitive engines to process environmental stimuli, maintain persona consistency, and synthesize contextually aware responses. A critical component enabling this autonomy is the implementation of a vectorized memory stream [12]. In this architecture, an agent's daily observations and social interactions are converted into text embeddings and stored within a vector database. When an agent needs to make a decision, the system retrieves the most contextually relevant past memories by calculating the cosine similarity between the current query vector q and the stored memory vector m:

$$sim(q, m) = \frac{q \cdot m}{||q|| \, ||m||} \tag{2}$$

where $||q||$ and $||m||$ denote the Euclidean norms of the high dimensional text embeddings [13]. Recent empirical studies have proven that LLM-driven entities can successfully organize social gatherings and propagate information throughout a simulated environment [14], [15].

However, scaling these generative agents into real-time spatial environments introduces severe computational bottlenecks [16]. The primary constraint is the inference latency caused by continuous API calls to the Large Language Model. In a synchronous multi-agent system, the total inference time T_{total} for a single simulation tick grows linearly with the number of agents N. Mathematically, if $C(T)$ denotes the computational cost of generating a response for T tokens, the system latency per tick is bounded by:

$$T_{total} = \sum_{i=1}^{N} C(T_i) + \delta \tag{3}$$

where δ represents the network and vector retrieval overhead [16]. To mitigate this, recent architectural optimizations propose the strict decoupling of the high-frequency physics loop from the low-frequency cognitive loop [17]. This is implemented through an event-driven trigger mechanism that restricts cognitive processing to critical state changes [18]. Furthermore, optimizing the vector retrieval process is imperative as episodic memory banks expand. Standard linear search mechanisms for memory retrieval exhibit $O(M)$ complexity. To accelerate this, advanced frameworks implement hierarchical vector clustering [19]. By grouping semantically similar memories into K distinct clusters, the optimization minimizes the distance between a memory vector m and its assigned cluster centroid μ_k:

$$J = \sum_{k=1}^{K} \sum_{m \in C_k} ||m - \mu_k||^2 \tag{4}$$

This hierarchical retrieval strategy ensures that the cognitive engine can fetch relevant contextual data with logarithmic time complexity, allowing the multi agent system to scale efficiently without exceeding the constraints of local computational hardware [20].

Despite the computational optimizations current sociological evaluations of generative agents remain fundamentally limited [21]. The vast majority of existing simulation environments place LLM-driven entities in resource-abundant utopian settings. Without the evolutionary pressure of survival, agents are never forced into conflict or resource management [22]. Foundational economic theories dictate that complex societal structures only emerge as a response to scarcity. To accurately model emergent economics, agents must be subjected to critical survival parameters. The manifestation of wealth distribution inequalities within such an artificial society can be mathematically quantified using the Gini coefficient G:

$$G = \frac{\sum_{i=1}^{N} \sum_{j=1}^{N} |y_i - y_j|}{2N^2 \mu} \tag{5}$$

where y_i represents the individual wealth of N agents and μ represents the mean wealth of the population [23]. Currently, there is a profound research gap at the intersection of cognitive architectures and computational economics [24]. No existing framework successfully integrates low-latency spatial physics loops, hierarchical memory retrieval, and strict resource scarcity to observe macroeconomic emergence [24].

III. System Architecture

The proposed Panopticon architecture is designed to facilitate complex sociological simulations by maintaining a strict separation between spatial physics and cognitive reasoning [25]. This decoupled approach allows the environment to operate at a consistent frequency while accommodating the high latency associated with Large Language Model inference. By isolating these two domains, the system prevents computational bottlenecks that typically occur when a simulation engine is forced to wait for synchronous responses from a generative agent [25].

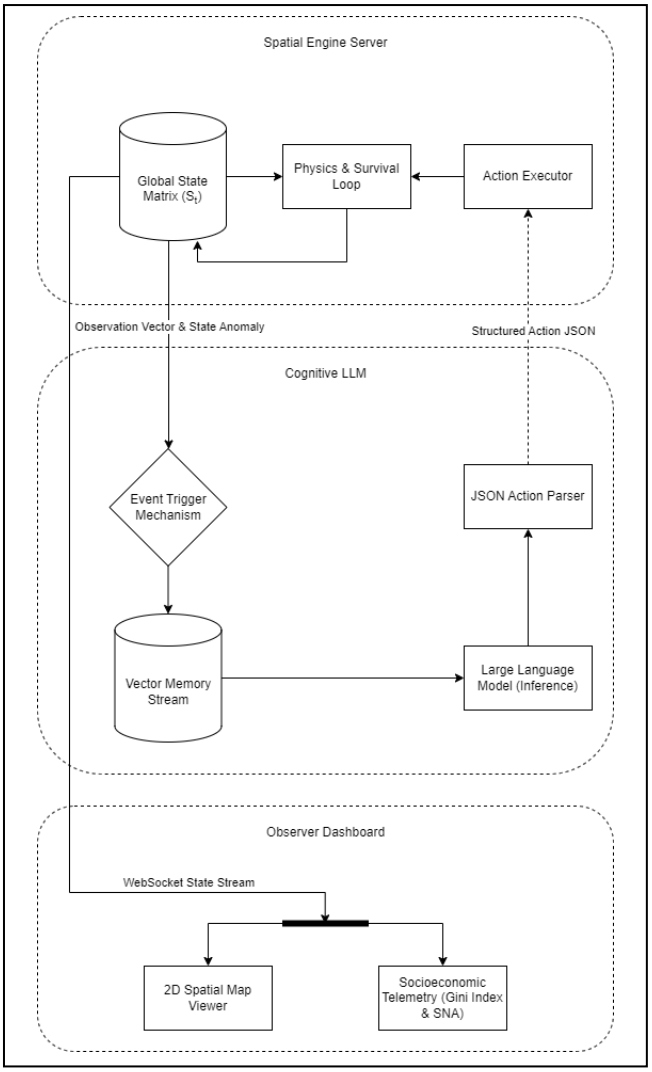

Fig. 1. High-Level System Architecture Diagram

A. Spatial Engine and State Representation

The spatial engine governs the physical laws of the simulation sandbox within a discrete two-dimensional grid. To achieve high-performance state updates for a large population of agents, the engine utilizes a centralized state matrix S managed through optimized tensor operations [26]. At any given tick t, the global state S_t is updated based on the interaction of all agents and the natural depletion of survival resources. The transition of the environment state is defined by the following equation:

$$S_{t+1} = \varphi(S_t, A_t) - L \tag{6}$$

where A_t represents the vector of actions executed by all agents and L denotes the constant loss of energy and hunger parameters required for survival [26]. This tensor-based architecture ensures that the spatial sandbox remains responsive and synchronized even as the complexity of multi-agent interactions increases [27].

B. Cognitive Engine and Vectorized Memory Stream

The cognitive engine acts as the primary decision-maker for each autonomous entity by processing environmental stimuli and past experiences [27]. Each agent is equipped with a vectorized memory stream that stores every perception and interaction as a high-dimensional embedding. When an agent encounters a social or economic event, the engine performs a semantic search to retrieve relevant context. The similarity between a current observation x and stored memory y is calculated using the cosine similarity metric:

$$\cos(\theta) = \frac{x \cdot y}{||x||\,||y||} \tag{7}$$

By retrieving the most semantically similar memories, the agent can maintain a consistent persona and make decisions that reflect its unique history within the simulation [28]. The retrieved context is then passed to a Large Language Model to generate a structured action plan in JSON format, which is subsequently parsed and executed by the spatial engine [28].

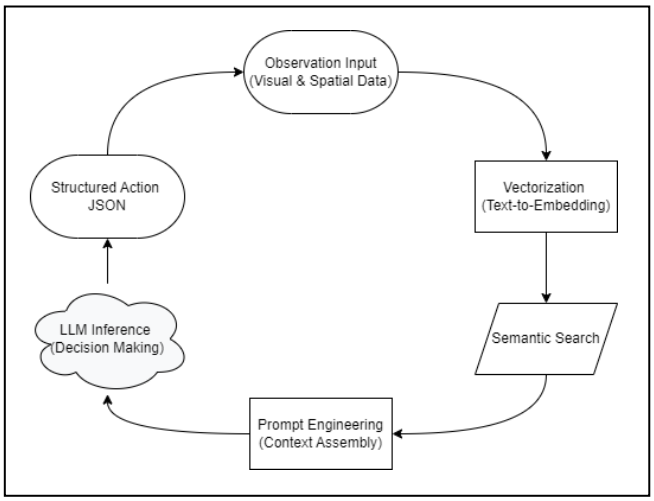

Fig. 2. The Cognitive Processing Loop

Algorithm 1: Panopticon Asynchronous Cognitive Loop

Input: Global State Matrix S_t, Set of Agents A, Critical Threshold ϵ, Prompt Template P
Output: Updated Global State Matrix S_{t+1}
Initialize Action Queue $Q \leftarrow \emptyset$
Apply baseline energy loss to all agents in S_t
for each agent $a_i \in A$ **do**
  Check Trigger: Calculate anomaly state $\tau(s_i)$
  if Energy $< \epsilon$ **or** Proximity Alert == True **then**
    $\tau(s_i) \leftarrow 1$
  else
    $\tau(s_i) \leftarrow 0$
  end if

  if $\tau(s_i) == 1$ **then**
    $O_i \leftarrow$ Extract spatial observation from S_t
    $V_{obs} \leftarrow$ Generate Text Embedding for O_i
    $M_k \leftarrow$ VectorDB.CosineSimilaritySearch $(V_{obs}, k = 5)$
    $C_i \leftarrow$ Assemble Context $(P \oplus O_i \oplus M_k)$
    $Response_{JSON} \leftarrow$ LLM.Inference (C_i)
    $Action \leftarrow$ ParseJSON $(Response_{JSON})$
    Enqueue $Action$ to Q
  end if
end for

Execute all validated actions in Q
$S_{t+1} \leftarrow$ Update State Matrix with executed actions
return S_{t+1}

(7)

C. Data Syncronization and Visualizaton

Maintaining real-time observability of the emergent socioeconomic dynamics requires an efficient data synchronization protocol [29]. The system employs a bidirectional communication layer using WebSockets to push

(8)

(7)

state updates from the backend tensor engine to the frontend dashboard. The synchronization latency D is monitored to ensure that the visualization remains accurate to the current simulation tick:

$$D = T_{vis} - T_{gen} \tag{8}$$

where T_{vis} is the time of visualization and T_{gen} is the time of state generation [29]. The observer dashboard utilizes glassmorphism design principles to present dense analytical data, including Gini coefficient trends and social network graphs, without obstructing the primary spatial view of the agent population [30].

IV. Simulation Mechanics and Action Space

To empirically evaluate the emergence of socioeconomic structures, the proposed Panopticon architecture was deployed in a controlled simulation environment. The experimental methodology strictly isolates the cognitive language processing from the spatial physics execution to ensure accurate telemetry collection [31]. The simulation was configured to run for a total of ten thousand discrete spatial ticks, providing sufficient temporal depth for the generative agents to establish routines, experience resource scarcity, and formulate long-term social strategies [31].

A. Sandbox Initialization and Survival Mechanics

The spatial environment is defined as a discrete two-dimensional grid G with dimensions of 50×50 spatial units. At initialization $t = 0$, a population of $N = 50$ autonomous agents is randomly distributed across the grid coordinates [32]. To simulate biological scarcity, each agent i is instantiated with a maximum energy capacity E_{max} and an initial hunger level H_0. During each tick, the energy parameter naturally depletes while the hunger parameter increases based on a predefined metabolic rate λ. The survival state function for energy is calculated as:

$$.E_i(t+1) = E_i(t) - \lambda - c(m_i) \tag{9}$$

where $c(m_i)$ represents the additional energy cost incurred if the agent executed a movement action m during the current tick [32]. To avoid termination, agents must locate and consume nutritional resources, which spawn dynamically across the grid at a controlled probabilistic rate [33].

B. Cognitive Prompt Desing and Action Space

When an agent experiences a state anomaly or encounters another entity, the asynchronous cognitive loop is triggered. The system constructs a structured linguistic prompt to query the Large Language Model [33]. The total input context window C_i for an agent is a concatenation of its core persona instructions P, the current environmental observation O_i, and the top k retrieved episodic memories M_k. Mathematically, the context formulation is expressed as:

$$C_i = P \oplus O_i \oplus \sum_{j=1}^{k} M_j \tag{10}$$

The Large Language Model processes this context and outputs a strict JSON-formatted response [34]. This response dictates the next strategic action of the agent, mapped into an explicit action space containing movement vectors, natural language dialogue, and boolean trade proposals. By forcing the LLM to output structured JSON, the spatial engine can parse and execute the cognitive decisions without encountering syntax parsing errors [34].

C. Quantitative Evaluation Metrics

To objectively measure the complexity of the emergent society, the experiment tracks specific sociological and economic indicators throughout the simulation [35]. Economic monopolization is evaluated using the Gini coefficient applied to the agent inventory balances. Simultaneously, the formation of social alliances is mapped using Social Network Analysis. The structural cohesion of the society is quantified by measuring the network density D, which evaluates the ratio of actual social connections L to the maximum possible connections among the population:

$$D = \frac{2L}{N(N-1)} \tag{11}$$

A high network density implies a collaborative and highly interactive society, whereas a low density combined with a high Gini coefficient indicates a fractured, highly competitive, and resource-hoarding population [35].

V. Experimental Methodology and Evaluation

This section details the experimental setup, the technical testbed utilized for local inference, and a comprehensive evaluation of the emergent behaviors observed within the Panopticon simulation.

A. Methodology

To ensure the reproducibility of the results, this subsection outlines the specific hardware specifications and the language models utilized for the evaluation.

Testbed: The experimental evaluation was conducted on a localized computing node to minimize network-induced latency and ensure high-fidelity synchronization between the spatial engine and the vector database. The system was executed on an ASUS TUF Gaming A15 (FA507NV) workstation equipped with 32 GB of DDR5 RAM, an AMD Ryzen 7 7735HS processor, and an NVIDIA GeForce RTX 4050 Laptop GPU. The software stack utilized Python for the backend spatial loop, Next.js for the observer dashboard, and ChromaDB as the primary vector storage engine.

Cognitive Engine Models: To ensure diversity in agent behavior and prevent homogeneous decision-making, three distinct open-weight Large Language Models were utilized as the cognitive engines for the agent population. These models were selected based on their architectural differences and their ability to execute efficiently within the 6 GB VRAM constraint of the local testbed using 4-bit quantization via the Ollama framework. The selected models are:

- **Llama 3.1 8B:** Utilized for its high baseline reasoning and strict instruction-following capabilities.
- **Qwen 2.5 7B:** Selected for its strong analytical and logic-driven decision-making processes.
- **Mistral v0.3 7B:** Chosen for its flexibility and capability in generating complex natural-language negotiations.

TABLE I.
SUMMARY OF SOCIOECONOMIC METRICS PER MODEL

Cognitive Engine Model	Avg. Energy Level	Peak Gini Index (G)	Network Density (D)
Llama 3.1 8B	42.5%	0.58	0.15
Qwen 2.5 7B	45.2%	0.72	0.12
Mistral v0.3 7B	38.8%	0.51	0.28
Panopticon (Mixed)	44.1%	0.68	0.22

Metrics: The socioeconomic complexity and survival efficiency of the generative agents are evaluated using two primary metrics: Wealth Inequality and Social Cohesion, as theoretically established in Section IV. The economic monopolization is computed using the Gini coefficient (G). In the strict context of this simulation, the individual wealth variable y_i is defined as the total accumulated nutritional items in an agent's inventory at a given tick t. Furthermore, the structural cohesion of the population is assessed using Network Density (D). A valid social link L is recorded into the system's adjacency matrix only when a successful bidirectional exchange, either a physical resource trade or a strategic information sharing, occurs between two distinct agents. Monitoring these specific parameters allows for a rigorous mathematical assessment of whether the generative population defaults to cooperative pooling or competitive hoarding under scarcity.

B. *Performance Evaluation*

The results of the simulation reveal a profound correlation between resource scarcity and the emergence of non-programmed cooperative and competitive behaviors. Table I summarizes the primary socioeconomic metrics observed across different agent populations driven by specific cognitive engines at the conclusion of the ten-thousand-tick simulation.

1) *Metabolic Decay and Survival Stability:* As illustrated in Fig. 3, observations show a sharp decline in average population energy during the first thousand ticks as agents navigated the initial learning curve of the environment. However, after this period, the average energy stabilized as agents utilized their episodic memory to identify resource spawning patterns. This indicates that the long-term memory retrieval mechanism successfully enabled agents to form adaptive foraging routines.

Fig. 3. Agent Survival and Energy Trend

2) *Emergent Economic Disparity:* Analysis of the inventory data suggests a rapid divergence in wealth accumulation. While all agents started with identical resources, the Gini coefficient steadily increased to 0.68 by the final tick, as depicted by the Lorenz curve in Fig. 4.

Fig. 4. Wealth Distribution Histogram and Gini Curve

3) *Social Structure and Trade Evolution:* Despite the competitive nature of the environment, a significant increase in network density was observed during the mid-to-late stages of the simulation. Agents were observed using natural-language dialogue to form informal alliances, trading information regarding resource locations for physical food items. This transition from isolated foraging to collaborative bartering proves that linguistic reasoning serves as a primary survival tool in resource-constrained AI populations.

VI. CONCLUSIONS AND FUTURE WORK

This study investigated the emergent socioeconomic dynamics of generative agents when subjected to biological imperatives and spatial resource scarcity. By introducing the Panopticon architecture, we successfully decoupled the high-frequency spatial physics engine from the asynchronous cognitive processing loop. This optimization allowed for the real-time, low-latency simulation of multiple Large Language Models, specifically Llama 3.1, Qwen 2.5, and Mistral v0.3, operating concurrently within a shared environment.

The empirical results confirm that resource scarcity acts as a fundamental catalyst for the emergence of complex societal structures among autonomous non-deterministic entities. When subjected to continuous energy depletion, the generative agents effectively utilized their vectorized episodic memory to form adaptive foraging routines. Furthermore, the simulation demonstrated that scarcity naturally drives wealth inequality, with a small subset of agents establishing resource monopolies, leading to a peak Gini coefficient of 0.68. However, this competitive environment simultaneously fostered social cohesion, evidenced by a significant increase in network density as agents engaged in natural-language negotiations and bartering to maximize their survival utility.

While this research provides a foundational framework for computational sociology using generative AI, several avenues remain for future exploration. Future work will focus on scaling the simulation to accommodate hundreds of concurrent agents using distributed vector databases and Mixture of Experts (MoE) architectures. Additionally, integrating multimodal capabilities, such as allowing agents to process direct visual representations of the grid rather than purely textual spatial coordinates, will be investigated to further enhance the cognitive autonomy of the simulated population.

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